
Research Interests

Efficient and trustworthy NLP, especially on speech separation, large language model (LLM) inference, retrieval-augmented generation (RAG), and computer agents.

Embodied intelligence, continuously extending to vision-language-action (VLA) models, world models, and reinforcement learning (RL).

Education

- Sep 2019–Jul 2024 **Ph.D. in Computer Science and Engineering**,
The Hong Kong University of Science and Technology, Kowloon, Hong Kong
Dissertation: “[Towards Private and Efficient Cross-Device Federated Learning](#)”
- Jul 2018 – Oct 2018 **Research Intern in Prof. Dean Tullsen’s Research Group**,
University of California, San Diego, La Jolla, US
○ [Defense](#) against Return-Oriented Programming with Context-Sensitive Decoding on x86-64.
- Sep 2015–Jun 2019 **B.Eng. in Computer Science**,
Zhejiang University, Hangzhou, China
GPA: 3.97/4.0, Graduated with Outstanding Honor (Zhejiang Province)

Experiences

- Nov 2024 – Present **Research Scientist in Central Software Institute**,
Huawei Technologies, Shenzhen, China
○ On-Device Audio Separation
- Speaker separation with and without voiceprint prior
- Target speech hearing with model-based beamforming
○ Efficient Inference for LLMs
- Decode in parallel for autoregressive LLMs
- Accelerate diffusion LLMs (dLLMs)
○ (L)LM Applications: RAG and Agents
- On-device anti-fraud detection by semantics for web browsing
- Large-scale RAG-based question answering
- A general-purpose, command-line agent compatible with MCP ([Hey](#))
- Sep 2019 – Jul 2024 **PhD Candidate in Computer Science and Engineering**,
The Hong Kong University of Science and Technology, Kowloon, Hong Kong
○ Private Inference for LLMs
- Protect system prompt with minimized impact on conversational quality ([PromptKeeper](#))
- Replicate membership inference and data reconstruction attacks against LLMs ([Tutorial](#))
○ Private and Efficient Federated Training
- Secure participant selection against adversarial servers ([Lotto](#))
- Enable dropout-resilient differential privacy without loss in time performance ([Dordis](#))
- Train asynchronously with principles for end-to-end speedup ([Pisces](#))

Publications

Conference and Journal Publications

- 2026 Jiayi Zhang, Zuo Gan, Chen Chen, **Zhifeng Jiang**, Hao Wang, Yifei Zhu, Quan Chen, Minyi Guo. “Boosting Gradient-based Training Diagnosis for Efficient and Accurate Federated Learning” , *accepted to appear in IEEE TMC (IF: 9.2, top journal in Mobile Computing)*

- 2025 **Zhifeng Jiang**, Zihua Jin, Guoliang He. “[PromptKeeper: Safeguarding System Prompts for LLMs](#)” , in the *Findings of the Association for Computational Linguistics: EMNLP 2025* (acceptance rate: 39%)
- 2024 **Zhifeng Jiang**, Peng Ye, Shiqi He, Wei Wang, Ruichuan Chen, Bo Li. “[Lotto: Secure Participant Selection against Adversarial Servers in Federated Learning](#)” , in the *Proc. of USENIX Security 2024* (acceptance rate: 17%)
- 2024 **Zhifeng Jiang**, Wei Wang, Ruichuan Chen. “[Dordis: Efficient Federated Learning with Dropout-Resilient Differential Privacy](#)” , in the *Proc. of ACM EuroSys 2024* (acceptance ratio: 15%)
- 2024 Peng Ye, **Zhifeng Jiang**, Wei Wang, Bo Li, Baochun Li. “[Feature Reconstruction Attacks and Countermeasures of DNN Training in Vertical Federated Learning](#)” , accepted to appear in *IEEE TDSC (IF: 7.5, top journal in Computer Security)*
- 2024 Yongkang Zhang, Haoxuan Yu, Chenxia Han, Cheng Wang, Baotong Lu, Yunzhe Li, **Zhifeng Jiang**, Yang Li, Xiaowen Chu, Huaicheng Li. “[SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs](#)” , accepted to appear in *ACM PPOPP 2025* (acceptance rate: 20%)
- 2024 Na Lv, Zhi Shen, Chen Chen, **Zhifeng Jiang**, Jiayi Zhang, Quan Chen, Minyi Guo. “[FedCA: Efficient Federated Learning with Client Autonomy](#)” , in the *Proc. of ICPP 2024* (acceptance rate: 29%)
- 2023 **Zhifeng Jiang**, Wei Wang, Bo Li, Qiang Yang. “[Towards Efficient Synchronous Federated Training: A Survey on System Optimization Strategies](#)” , in *IEEE TBD, Volume 9, Issue 2 (IF: 5.7, top journal in Big Data)*
- 2022 **Zhifeng Jiang**, Wei Wang, Baochun Li, Bo Li. “[Pisces: Efficient Federated Learning via Guided Asynchronous Training](#)” , in the *Proc. of ACM SoCC 2022* (acceptance ratio: 25%)
- 2021 Minchen Yu, **Zhifeng Jiang**, Hok Chun Ng, Wei Wang, Ruichuan Chen, Bo Li. “[Gillis: Serving Large Neural Networks in Serverless Functions with Automatic Model Partitioning](#)” , in the *Proc. of IEEE ICDCS 2021* (acceptance ratio: 20%; **Best Paper Runner-Up**, 3 out of 97 accepted submissions)
- Manuscripts**
- 2021 **Zhifeng Jiang**, Wei Wang, Yang Liu. “[FLASHE: Additively Symmetric Homomorphic Encryption for Cross-Silo Federated Learning](#)” , in *arXiv preprint (Citation: 87+)*

Honors and Awards

- 2024, 2023 Redbird Academic Excellence Award, HKUST
- 2021 Best Paper Runner-Up Award (Top 3 out of 97 accepted papers), IEEE ICDCS
- 2019 Outstanding Graduate Award (Top 1%), Zhejiang Province
- 2017 He Zhijun Scholarship (Top 10 in Dept. of CS), ZJU
- 2017 National Scholarship (Top 0.1% nationwide), Ministry of Education, China

Professional Service

- Journal Reviewer [IEEE Transactions on Mobile Computing](#), [IEEE Transactions on Big Data](#), [IEEE Transactions on Neural Networks and Learning Systems](#), [IEEE Transactions on Knowledge and Data Engineering](#), [IEEE Transactions on Dependable and Secure Computing](#), [IEEE Transactions on Computers](#).
- Conference Reviewer [ICLR 2026 Workshop RSI](#), [Shadow ACM EuroSys 2023](#).
- AEC Member [USENIX OSDI 2022](#), [USENIX ATC 2022](#), [ACM SOSP 2021](#).

Journal IEEE Transactions on Network Science and Engineering, IEEE Transactions on Depend-
Sub-Reviewer able and Secure Computing, IEEE Transactions on Big Data.
Conference IEEE INFOCOM 2020-2024, IEEE ICDCS 2024, 2023 and 2021, IEEE/ACM IWQoS
Sub-Reviewer 2020-2021, IEEE WoWMoM 2021, IEEE ICNP 2020.

Teaching

Teaching Assistant HKUST COMP3511 Operating System: Fall 2022, Fall 2020.
HKUST COMP4651 Cloud Computing: Fall 2021.
HKUST COMP4521 Mobile Application Development: Spring 2020.
ZJU Operating System (Educational Reform Class): Fall 2018.